

JDK & JRE

1) How java works internally?

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Java works in two phases

1) Compiler phase

- The programmer writes source code in a .java file.
- Java compiler (javac) checks syntax errors.
- compiler converts source code into bytecode (.class file).

2) Running / Execution phase

- It is done by interpreter which needs one environment which is provided by JVM (Java Virtual Machine)
- Here, it converts byte code into machine code

2) What is JDK and JRE?

→ JDK (Java Development kit)

- JDK is used for developing java applications.

It contains :

- JRE

- compiler

JRE (Java Runtime environment)

JRE is used only to run Java applications.

It contains:

- JVM

3] What is role of compiler & interpreter?

→ Compiler

- It converts source code (.java) into bytecode (.class)
- It checks syntax errors

Interpreter:

- It converts bytecode into machine code.
- Mlc code is understood by computer

4] What is JVM?

→ JVM is a virtual machine that runs Java bytecode.

- It converts bytecode into machine code.
- provides platform independence.

5] Why Java is called platform independent?

→ Because,

- Java source code is converted into bytecode.
- Bytecode can run on any OS having JVM.

6] Why create Main method in Test class?

→ Main method is starting point of java program.

- JVM starts execution from main methods.
- Without main method, program execution can't start.